R2

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1 New Code/Dev stuff

- (maybe) Setup `BH-Sol_Filter.py` to use multiprocessing, cutting time down for filtering into the BH-Sol candidates massively.
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2 Questions/Theories/Ponderings

2.1 PN Detection when looking for BH-Sol systems

In "An upper limit on the frequency of short-period black hole companions to Sun-like stars" they looked solar for tidal deformations from another orbiting BH on the sunlike star. However, wouldn't it also make sense to factor in the fact that said systems *must* have evolved through CE evolution and ejected a shell? Thus producing some sort of PN ring? I'll have to look into it more, as I'd imagine said ring would also be ionized for a brief(?) period of time before then largely acting to obscure (and maybe redden) the sun-like star

2.2 POSYDON Dataset to Classify Tertiary System

Couldn't one theoretically use a large POSYDON population and its properties to then compare to the properties of observed systems to determine if they evolved through binary evolution? Accidentally stumbled across this around a year ago with V404 Cygni, but it feels relatively feasible to make a grouping algorithm to find systems which are predicted to have evolved through tertiary means (or capture/dynamic methods)

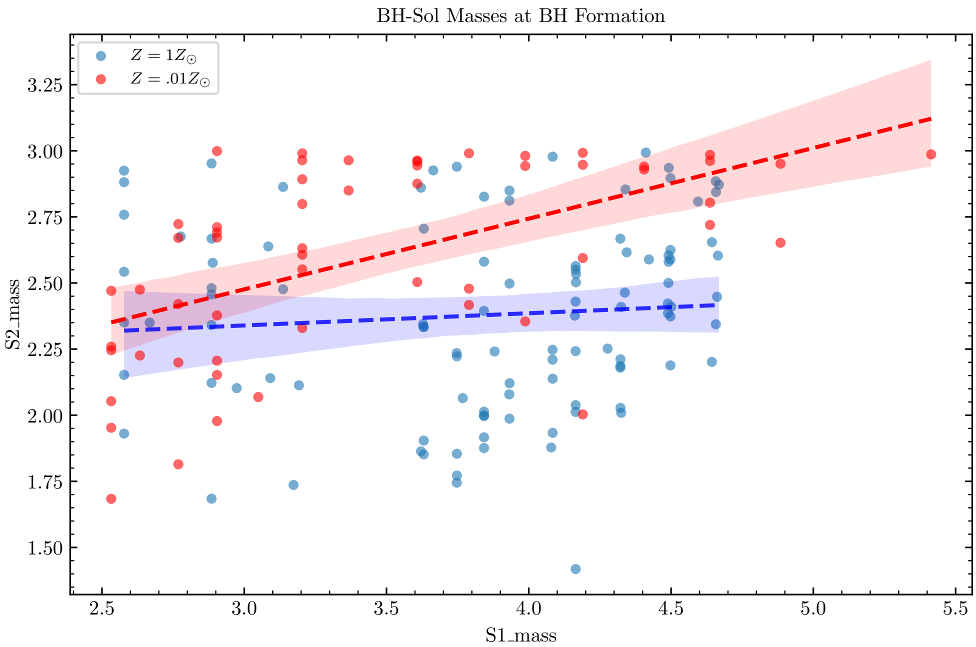
2.3 Striation in Mass Loss Graphs

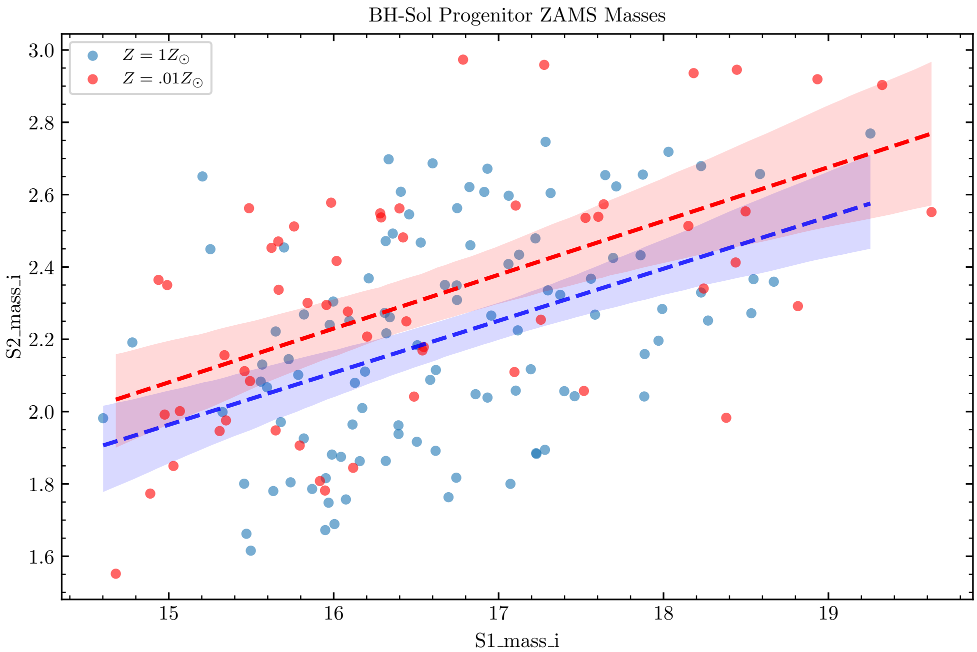
In the graphs with ΔM_{SNe} we see some striation happening with masses loses between 3.5 to 3.7 and then again between 3.9 to 4.1, is this just a byproduct of POSYDON resolution or is there some underlying physics concept?

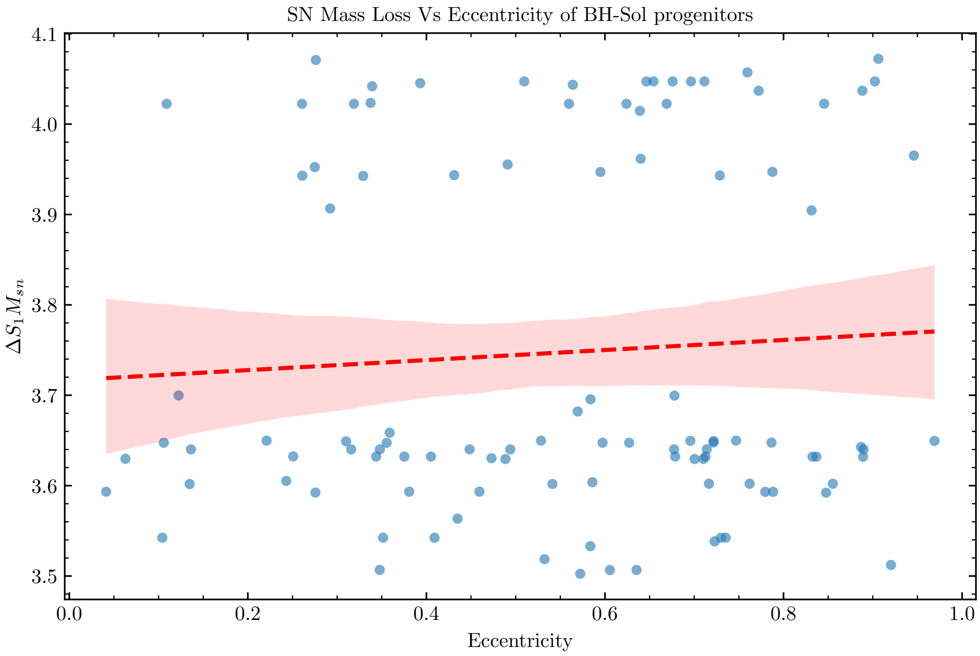
2.4 Distinct Resolution Scale

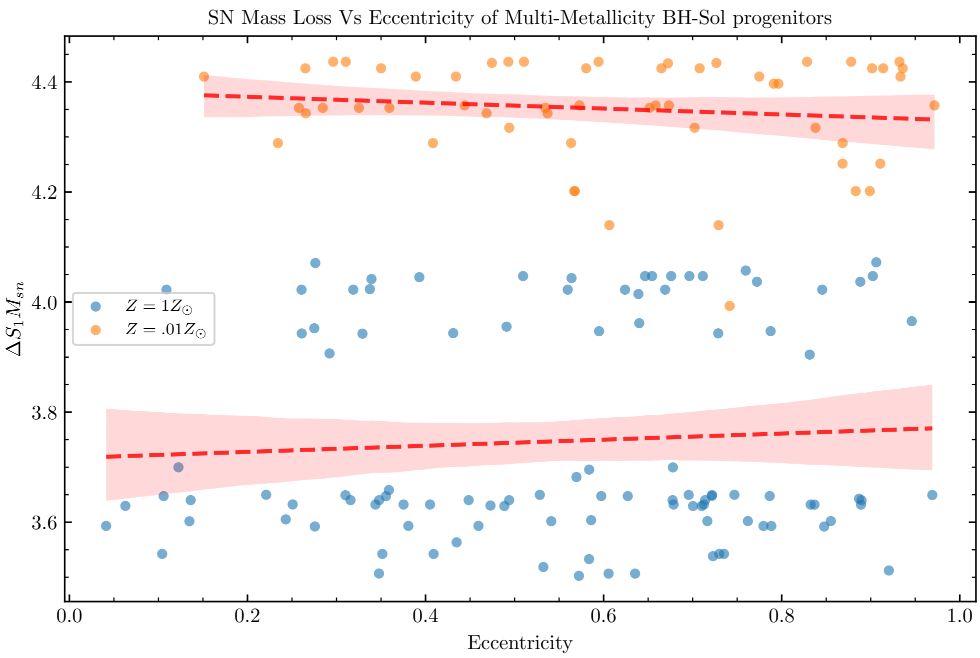
In the BH-Sol Masses at BH Formation graph **??** we can see the effects of the resolution of the POSYDON reference grids (could also possibly be some sort of underlying physics), also, notably, there doesn't seem to be a massive difference between the lower resolution one ($.0Z_{\odot}$) and the higher res.

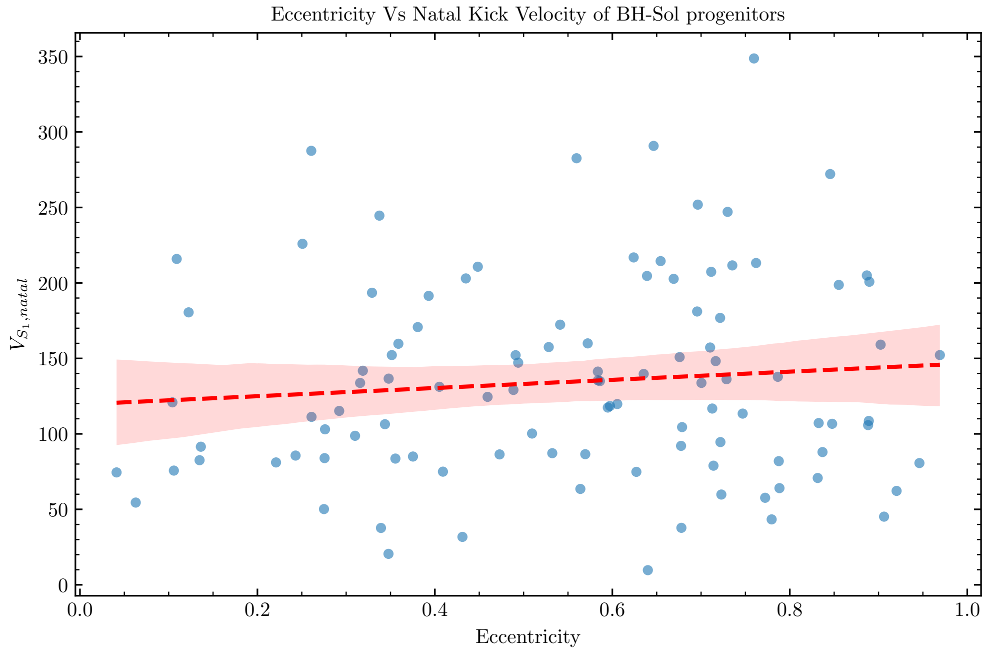
3 10 Mill Pop Graphs





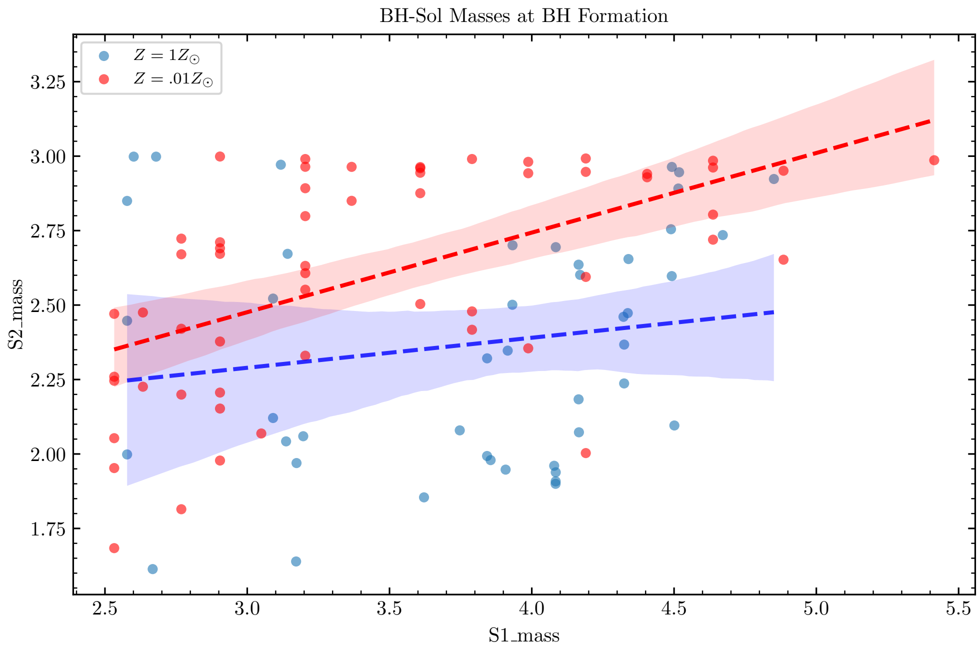

SN Mass Loss Vs Eccentricity of BH-Sol progenitors
$\Delta S_1 M_{sn}$
Eccentricity
4.1
4.0
3.9
3.8
3.7
3.6
3.5
0.0
0.2
0.4
0.6
0.8
1.0

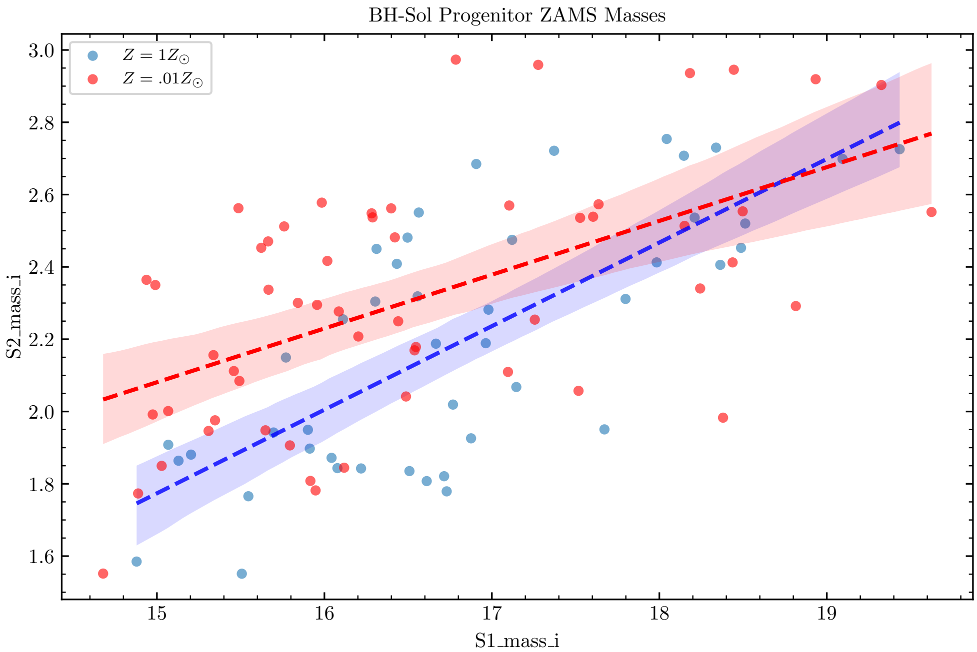

SN Mass Loss Vs Eccentricity of Multi-Metallicity BH-Sol progenitors
$\Delta S_1 M_{sn}$
Eccentricity
$Z = 1Z_{\odot}$
$Z = .01Z_{\odot}$
4.4
4.2
4.0
3.8
3.6
0.0
0.2
0.4
0.6
0.8
1.0

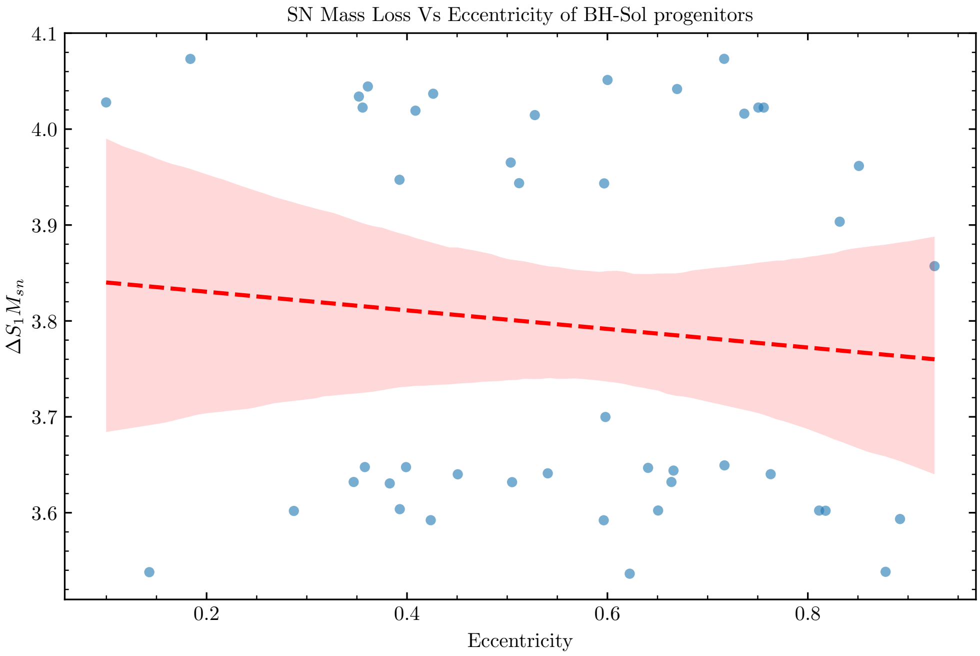


Eccentricity Vs Natal Kick Velocity of BH-Sol progenitors
$V_{S_1,natal}$
350
300
250
200
150
100
50
0
0.0
0.2
0.4
0.6
0.8
1.0
Eccentricity

4 1 Mill Graphs







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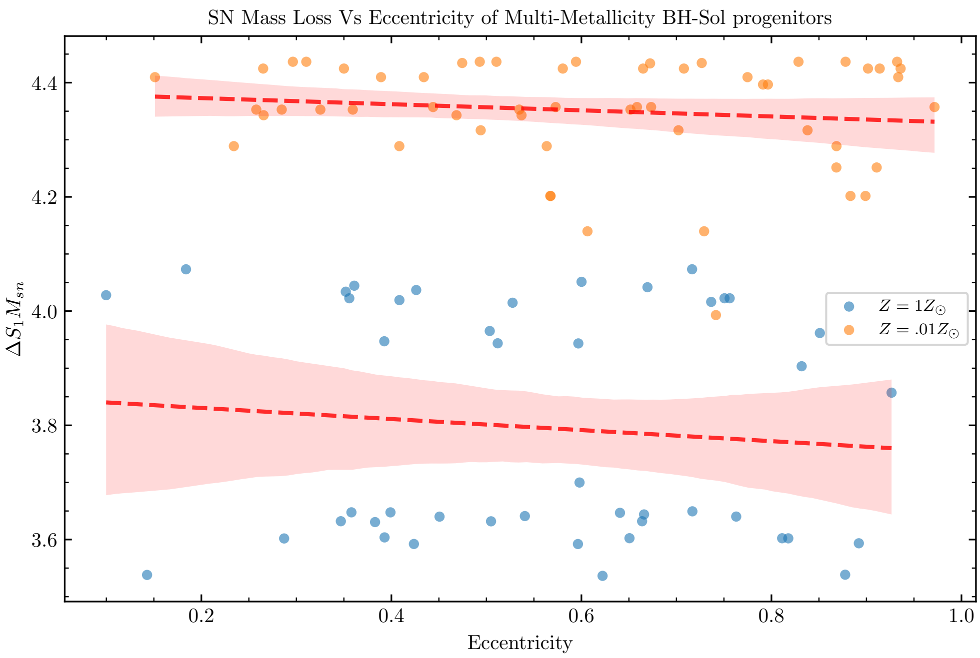

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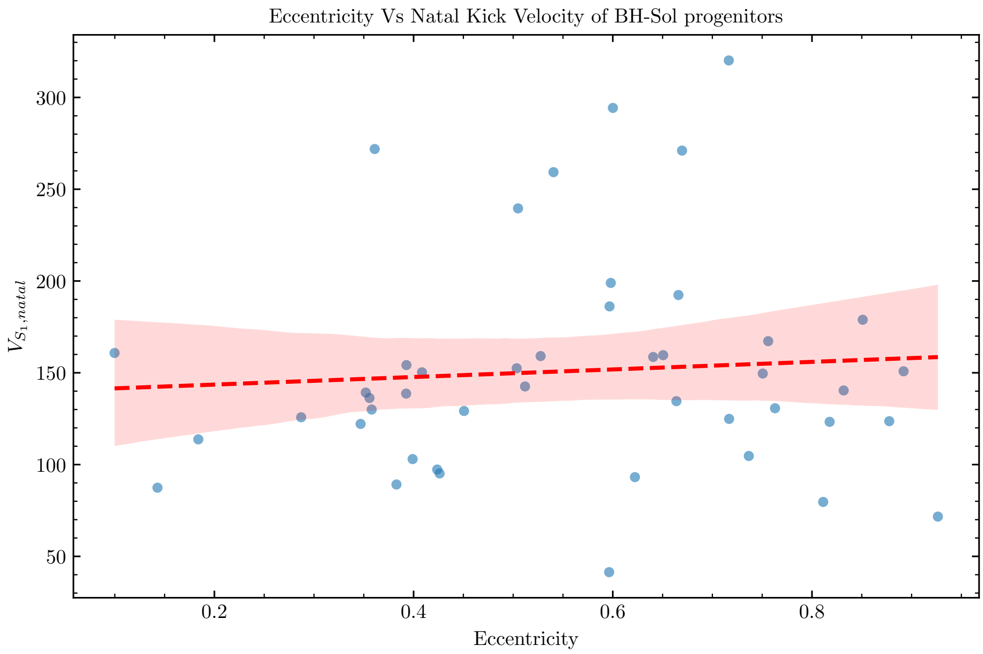
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